Nexus Architecture Atlas

This walkthrough maps Nexus's durable user flows to their owning components.

User flow	Primary owner	Supporting contract	Evidence
Install or repair media	`scripts/installer/`	Owner-aware lease and atomic runtime state	Installer tests plus packaged operator run
Generate an image	`desktop/sidecar/src/diffusion/dispatcher.ts` and Python pipelines	Shared diffusion readiness API	PNG validator and retained artifact
Edit or recover a saved image	`desktop/sidecar/src/studio/imageEditRuntime.ts` and the Image Studio viewer	Revisioned recipes, approval checkpoints and viewer recovery; shared originals retained	Approval, retry, restart, restore and revocation tests
Generate a video	`desktop/sidecar/src/diffusion/videoDispatcher.ts` and Python pipelines	Shared diffusion readiness API	Video probe and retained artifact
Browse and install models	`core/registry/` and Settings	Shared availability/recommendation tuple	Cross-language rank fixture
Read chat usage	`core/chat/` and shared chat UI	Role-local token provenance	Protocol, hydration, and UI tests
Organize chats and work across folders	Chat explorer, Coding page and workspace scope	Persisted folder colors; union-of-roots path boundary	Database reopen, picker, persistence and denial tests

End-to-end path

1. Catalog-source reachability checks and desktop payload preparation are packaging prerequisites. The installer provisions the media runtime.
2. The sidecar reads the atomic readiness record and exposes one repair contract to Image Studio and Video Lab.

3. Generation jobs check the selected model directory layout before loading models and finalize artifacts before reporting success.
4. The desktop renders truthful capability, model, transcript, archive, and workspace state from shared contracts.
5. Packaged acceptance requires installing the exact candidate, retaining the installer-to-payload binding, and exercising the installed runtime with the media smoke harness. The harness alone does not establish that binding.

Release-specific logs and test counts live in the corresponding release or archive tree.